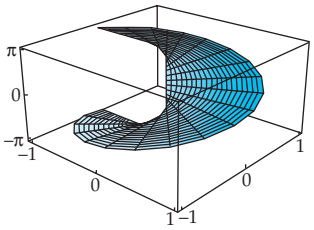



**(a)** The cut disk $A_0$




**(b)** $A_0$ placed in $xyz$-space

**FIGURE 2.4.18** The cut disk $A_0$

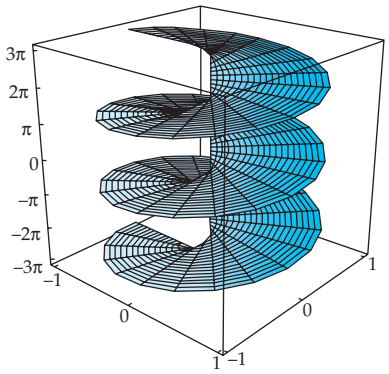



**FIGURE 2.4.19** The Riemann surface for $G(z) = \arg(z)$

## EXERCISES 2.4

Answers to selected odd-numbered problems begin on page ANS-8.

### 2.4.1 The Power Function $z^n$

In Problems 1–14, find the image of the given set under the mapping $w = z^2$. Represent the mapping by drawing the set and its image.

**1.** the ray $\arg(z) = \dfrac{\pi}{3}$

**2.** the ray $\arg(z) = -\dfrac{3\pi}{4}$

**3.** the line $x = 3$

**4.** the line $y = -5$

**5.** the line $y = -\dfrac{1}{4}$

**6.** the line $x = \dfrac{3}{2}$

**7.** the positive imaginary axis

**8.** the line $y = x$

**9.** the circular arc $|z| = \dfrac{1}{2}, 0 \le \arg(z) \le \pi$

**10.** the circular arc $|z| = \dfrac{4}{3}, -\dfrac{\pi}{2} \le \arg(z) \le \dfrac{\pi}{6}$

**11.** the triangle with vertices 0, 1, and $1 + i$

**12.** the triangle with vertices 0, $1 + 2i$, and $-1 + 2i$

**13.** the square with vertices 0, 1, $1 + i$, and $i$

**14.** the polygon with vertices 0, 1, $1 + i$, and $-1 + i$

In Problems 15–20, find the image of the given set under the given composition of a linear function with the squaring function.

**15.** the ray $\arg(z) = \dfrac{\pi}{3}$; $f(z) = 2z^2 + 1 - i$

**16.** the line segment from 0 to $-1+ i$; $f(z) = \sqrt{2}z^2 + 2 - i$

**17.** the line $x = 2$; $f(z) = iz^2 - 3$

**18.** the line $y = -3$; $f(z) = -z^2 + i$

**19.** the circular arc $|z| = 2, 0 \le \arg(z) \le \dfrac{\pi}{2}$; $f(z) = \frac{1}{4}e^{i\pi/4}z^2$

**20.** the triangle with vertices 0, 1, and $1 + i$; $f(z) = -\frac{1}{4}iz^2 + 1$

**21.** Find the image of the ray $\arg(z) = \pi/6$ under each of the following mappings.

**(a)** $f(z) = z^3$ **(b)** $f(z) = z^4$ **(c)** $f(z) = z^5$

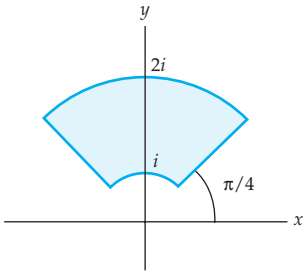



**FIGURE 2.4.20** Figure for Problems 23 and 24

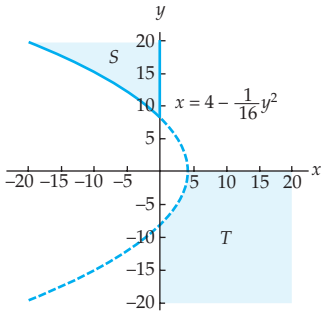



**FIGURE 2.4.21** Figure for Problem 39

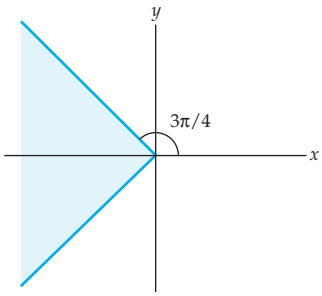



**FIGURE 2.4.22** Figure for Problem 40

**22.** Find the image of the first quadrant of the complex plane under each of the following mappings.

(a) $f(z) = z^2$ (b) $f(z) = z^3$ (c) $f(z) = z^4$

**23.** Find the image of the region $1 \le |z| \le 2$, $\pi/4 \le \arg(z) \le 3\pi/4$, shown in **FIGURE 2.4.20** under each of the following mappings.

(a) $f(z) = z^2$ (b) $f(z) = z^3$ (c) $f(z) = z^4$

**24.** Find the image of the region shown in Figure 2.4.20 under each of the following mappings.

(a) $f(z) = 3z^2 + i$ (b) $f(z) = (i+1)z^3 + 1$ (c) $f(z) = \frac{1}{2}z^4 - i$

### 2.4.2 The Power Function $z^{1/n}$

In Problems 25–30, use (13) to find the value of the given principal $n$th root function at the given value of $z$.

**25.** $z^{1/2}$, $z = -i$

**26.** $z^{1/2}$, $z = 2 + i$

**27.** $z^{1/3}$, $z = -1$

**28.** $z^{1/3}$, $z = -3 + 3i$

**29.** $z^{1/4}$, $z = -1 + \sqrt{3}i$

**30.** $z^{1/5}$, $z = -4\sqrt{3} + 4i$

In Problems 31–38, find the image of the given set under the principal square root mapping $w = z^{1/2}$. Represent the mapping by drawing the set and its image.

**31.** the ray $\arg(z) = \dfrac{\pi}{4}$

**32.** the ray $\arg(z) = -\dfrac{2\pi}{3}$

**33.** the positive imaginary axis

**34.** the negative real axis

**35.** the arc $|z| = 9$, $-\dfrac{\pi}{2} \le \arg(z) \le \pi$

**36.** the arc $|z| = \dfrac{4}{7}$, $-\dfrac{\pi}{2} \le \arg(z) \le \dfrac{\pi}{4}$

**37.** the parabola $x = \dfrac{9}{4} - \dfrac{y^2}{9}$

**38.** the parabola $x = \dfrac{y^2}{10} - \dfrac{5}{2}$

**39.** (a) Find the image of the region $S$ shown in **FIGURE 2.4.21** under the principal square root mapping $w = z^{1/2}$.

(b) Find the image of the region $T$ shown in Figure 2.4.21 under the principal square root mapping $w = z^{1/2}$.

**40.** Find the image of the region shown in **FIGURE 2.4.22** under the principal square root function $w = z^{1/2}$. (Be careful near the negative real axis!)

### Focus on Concepts

**41.** Use a procedure similar to that used in Example 2 to find the image of the hyperbola $xy = k$, $k \neq 0$, under $w = z^2$.

**42.** Use a procedure similar to that used in Example 2 to find the image of the hyperbola $x^2 - y^2 = k$, $k \neq 0$, under the mapping $w = z^2$.

**43.** Find two sets in the complex plane that are mapped onto the ray $\arg(w) = \pi/2$ by the function $w = z^2$.

**44.** Find two sets in the complex plane that are mapped onto the set bounded by the curves $u = -v$, $u = 1 - \frac{1}{4}v^2$, and the real axis $v = 0$ by the function $w = z^2$.

**45.** In Example 2 it was shown that the image of a vertical line $x = k$, $k \neq 0$, under $w = z^2$ is the parabola $u = k^2 - \dfrac{v^2}{4k^2}$. Use this result, your knowledge of linear mappings, and the fact that $w = -(iz)^2$ to prove that the image of a horizontal line $y = k$, $k \neq 0$, is the parabola $u = -\left(k^2 - \dfrac{v^2}{4k^2}\right)$.

**46.** Find three sets in the complex plane that map onto the set $\arg(w) = \pi$ under the mapping $w = z^3$.

**47.** Find four sets in the complex plane that map onto the circle $|w| = 4$ under the mapping $w = z^4$.

**48.** Find the image of the line $y = mx$ under the mapping $w = z^n$ for $n \geq 2$.

**49.** (**a**) Proceed as in Example 6 to show that the complex linear function $f(z) = az + b$, $a \neq 0$, is one-to-one on the entire complex plane.

(**b**) Find a formula for the inverse function of the function in (a).

**50.** (**a**) Proceed as in Example 6 to show that the complex function $f(z) = \dfrac{a}{z} + b$, $a \neq 0$, is one-to-one on the set $|z| > 0$.

(**b**) Find a formula for the inverse function of the function in (a).

**51.** Find the image of the half-plane $\text{Im}(z) \geq 0$ under each of the following principal $n$th root functions.

(**a**) $f(z) = z^{1/2}$ (**b**) $f(z) = z^{1/3}$ (**c**) $f(z) = z^{1/4}$

**52.** Find the image of the region $|z| \leq 8$, $\pi/2 \leq \arg(z) \leq 3\pi/4$, under each of the following principal $n$th root functions.

(**a**) $f(z) = z^{1/2}$ (**b**) $f(z) = z^{1/3}$ (**c**) $f(z) = z^{1/4}$

**53.** Find a function that maps the entire complex plane excluding 0 onto the set $2\pi/3 < \arg(w) \leq 4\pi/3$.

**54.** Read part (*ii*) of the *Remarks*, and then describe how to construct a Riemann surface for the function $f(z) = z^3$.

In Problems 55 and 56, (**a**) use mappings to determine upper and lower bounds on the modulus of the given function $f(z)$ defined on the given set $S$. That is, find real values $L$ and $M$ such that $L \leq |f(z)| \leq M$ for all $z$ in $S$ and (**b**) find complex values $z_0$ and $z_1$ in $S$ such that $|f(z_0)| = L$ and $|f(z_1)| = M$.

**55.** $f(z) = 2iz^2 - i$; $S$ is the quarter disk $|z| \leq 2$, $0 \leq \arg(z) \leq \pi/2$

**56.** $f(z) = \frac{1}{3}z^2 + 1 - i$; $S$ is the set defined by $2 \leq |z| \leq 3$, $0 \leq \arg(z) \leq \pi$

## 2.5 Reciprocal Function

***To The Instructor***: In this section, we study the complex mapping $w = 1/z$. This section can be skipped without affecting the development of topics in Chapters 3–6. However, topics presented in this section will be used in the study of conformal mappings in Chapter 7.

**INTRODUCTION** In Sections 2.3 and 2.4 we examined some special types of complex polynomial functions as mappings of the complex plane. Analogous to real functions, we define a **complex rational function** to be a function of the form $f(z) = p(z)/q(z)$ where both $p(z)$ and $q(z)$ are complex polynomial functions. In this section, we study the most basic complex rational function, the reciprocal function $1/z$, as a mapping of the complex plane. An important property of the reciprocal mapping is that it maps certain lines onto circles.

### Reciprocal Function

The function $1/z$, whose domain is the set of all nonzero complex numbers, is called the **reciprocal function**. To study the reciprocal function as a complex mapping $w = 1/z$, we begin by expressing this function in exponential notation. Given $z \neq 0$, if we set $z = re^{i\theta}$, then we obtain:

$$w = \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r}e^{-i\theta}. \tag{1}$$

From (1), we see that the modulus of $w$ is the reciprocal of the modulus of $z$ and that the argument of $w$ is the negative of the argument of $z$. Therefore, the reciprocal